6.8. The observed quadrupole moment of ^{59}Co is 0.40 barn. (a) What is the deformation value β? (b) What spin value is expected from the Nilsson diagram?

6.9. Which neutron and proton states account for the spin value I of ^{14}N?

6.10. A γ-line at 0.146 MeV is assigned to a $+4 \rightarrow +0$ rotational level change in ^{238}Pu. (a) What should the energy of the +2 and +6 rotational levels be? Compare with the measured values of 0.044 and 0.304 MeV. (b) If ^{238}Pu is considered to be a homogenous sphere, what will its apparent radius be? Compare with that obtained using (4.14).

6.11. A ^{239}Pu compound is placed in a test tube in a 40 MHz nmr machine. At what field strength does resonance occur with the nuclear spin? Is the measurement possible? Relevant data appear in Table 6.3.

6.12. Using the Gamow theory the probability for tunneling of an α-particle in the decay of ^{238}U is $1{:}10^{38}$, and the α-particle hits the walls about 10^{21} times per second. What average lifetime can be predicted for ^{238}U from this information?

6.13. Calculate the half-life for α-decay of ^{147}Sm assuming that Q_α is 2.314 MeV. Compare the result with the measured half-life and compute the hindrance factor.

6.9. LITERATURE

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CHAPTER 7

Absorption of Nuclear Radiation

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Our understanding of the nature of nuclear particles is based on their mode of interaction with matter. If the interaction of the various nuclear particles with various material is known it is possible to use this knowledge for e.g. proper utilization and construction of detection and measuring devices for radiation, the design of radiation shielding, the medical and biological applications of radiation, radiochemical synthesis, etc.

The term *nuclear radiation* is used to include all elementary particles, both uncharged (e.g. photons) and charged (e.g. electrons), having energies in excess of approximately 100 eV. This is valid whether the particles have been produced through nuclear reactions (spontaneous or induced) or have acquired their energy in electrostatic accelerators.

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